

```
(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B103-Lab1/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/CIP-B103-Lab1

(kali㉿kali)-[~/CIP-B103-Lab1]
$ pwd
/home/kali/CIP-B103-Lab1

(kali㉿kali)-[~/CIP-B103-Lab1]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

(kali㉿kali)-[~/CIP-B103-Lab1]
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease
Get:2 http://kali.download/kali kali-rolling/main amd64
Get:3 http://kali.download/kali kali-rolling/main amd64
Get:4 http://kali.download/kali kali-rolling/contrib am
Get:5 http://kali.download/kali kali-rolling/contrib am
Fetched 74.7 MB in 4min 36s (270 kB/s)
1736 packages can be upgraded. Run 'apt list --upgradab

(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo apt install -y apache2 curl wireshark tshark
apache2 is already the newest version (2.4.68-1).
curl is already the newest version (8.20.0-5).
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
The following packages were automatically installed and
  libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-2
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgradi
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service
install.
Executing: /usr/lib/systemd/systemd-sysv-install enable

(kali㉿kali)-[~/CIP-B103-Lab1]
$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.se
   Active: active (running) since Tue 2026-07-28 07:3
 Invocation: dce5043be1b5485aabe10a1bbf6f1305
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 966 (apache2)
    Status: "Total requests: 0; Idle/Busy workers 100/
    Tasks: 6 (limit: 2075)
   Memory: 21.1M (peak: 21.9M)
      CPU: 348ms
   CGroup: /system.slice/apache2.service
           └─ 966 /usr/sbin/apache2 -k start -DFOREGR
              1008 /usr/sbin/apache2 -k start -DFOREGR
              1009 /usr/sbin/apache2 -k start -DFOREGR
              1010 /usr/sbin/apache2 -k start -DFOREGR
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ printf '<!DOCTYPE html>\n<html><body><h1>CIP-B103 HTTP Evidence</h1><p>Name: Fuseini Imoru</p><p>Reg No: 17291</p></body></html>\n' | sudo tee /var/www/html/basic.html
<!DOCTYPE html>
<html><body><h1>CIP-B103 HTTP Evidence</h1><p>Name: Fuseini Imoru</p><p>Reg No: 17291</p></body></html>

(kali㉿kali)-[~/CIP-B103-Lab1]
$ curl -v http://127.0.0.1/basic.html
* Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 38024
* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 28 Jul 2026 11:50:40 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Tue, 28 Jul 2026 11:50:25 GMT
< ETag: "78-657aa71cb8292"
```

```
< Last-Modified: Tue, 28 Jul 2026 11:50:25 GMT
< ETag: "78-657aa71cb8292"
< Accept-Ranges: bytes
< Content-Length: 120
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>CIP-B103 HTTP Evidence</h1><p>Name: Fuseini Imoru</p><p>Reg No: 17291</p></body></h
tml>
* Connection #0 to host 127.0.0.1:80 left intact
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo tshark -i lo -w /tmp/basic.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
10 ^C
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ cp /tmp/basic.pcapng ~/CIP-B103-Lab1/evidence/
cp: cannot open '/tmp/basic.pcapng' for reading: Permis
sion denied
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ ls -l /tmp/basic.pcapng
-rw----- 1 root root 1812 Jul 28 07:56 /tmp/basic.pca
png
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo cp /tmp/basic.pcapng ~/CIP-B103-Lab1/evidence/
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo chown kali:kali ~/CIP-B103-Lab1/evidence/basic
.pcapng
```

```
zsh: corrupt history file /home/kali/.zsh_history
(kali㉿kali)-[~/CIP-B103-Lab1]
$ curl -v http://127.0.0.1/basic.html
* Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 8
0) from 127.0.0.1 port 39834
* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 28 Jul 2026 11:55:40 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Tue, 28 Jul 2026 11:50:25 GMT
< ETag: "78-657aa71cb8292"
< Accept-Ranges: bytes
< Content-Length: 120
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>CIP-B103 HTTP Evidence</h1><p>Name: Fus
eini Imoru</p><p>Reg No: 17291</p></body></html>
```



```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ ls -l ~/CIP-B103-Lab1/evidence/basic.pcapng
-rw-r--r-- 1 kali kali 1812 Jul 28 07:57 /home/kali/CIP-B103-Lab1/evidence/basic.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sha256sum evidence/basic.pcapng | tee reports/basic_pcap_sha256.txt
53d0cadf491d24b130821c786a5d4a9478eddd0a01986aeae0d1b36b2a647334f evidence/basic.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ tshark -r evidence/basic.pcapng -Y http
  4 0.000168092      127.0.0.1 → 127.0.0.1      HTTP 149
GET /basic.html HTTP/1.1
  6 0.001945901      127.0.0.1 → 127.0.0.1      HTTP 437
HTTP/1.1 200 OK (text/html)
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ curl -v http://127.0.0.1/basic.html 2>&1 | tee reports/curl_verbose.txt
* Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 38292
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0     0      0     0      0      0     0  0* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 28 Jul 2026 12:10:18 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Tue, 28 Jul 2026 11:50:25 GMT
< ETag: "78-657aa71cb8292"
< Accept-Ranges: bytes
< Content-Length: 120
< Vary: Accept-Encoding
< Content-Type: text/html
```

```

<
{ [120 bytes data]
100 120 100 120 0 0 62337 0 0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>CIP-B103 HTTP Evidence</h1><p>Name: Fuseini Imoru</p><p>Reg No: 17291</p></body></
tml>

(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-28 07:35:31 EDT; 35min ago
  Invocation: dce5043be1b5485aabe10a1bbf6f1305
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 966 (apache2)
      Status: "Total requests: 3; Idle/Busy workers 100/0;Requests/sec: 0.00143; Bytes served/sec: 0 B/sec"
    Tasks: 6 (limit: 2075)
  Memory: 16.7M (peak: 21.9M, swap: 2.8M, swap peak: 2.8M)
    CPU: 501ms
  CGroup: /system.slice/apache2.service
          └─ 966 /usr/sbin/apache2 -k start -DFOREGROUND

```

```

└─1008 /usr/sbin/apache2 -k start -DFOREGROUND
└─1009 /usr/sbin/apache2 -k start -DFOREGROUND
└─1010 /usr/sbin/apache2 -k start -DFOREGROUND
└─1011 /usr/sbin/apache2 -k start -DFOREGROUND
└─1012 /usr/sbin/apache2 -k start -DFOREGROUND

Jul 28 07:35:31 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Jul 28 07:35:31 kali apachectl[966]: AH00558: apache2: Could not reliably determine the serv...essage
Jul 28 07:35:31 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Hint: Some lines were ellipsized, use -l to show in full.

(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo ss -lntp | grep ':80'
LISTEN 0      511          *:80          *: * users:(("apache2",pid=1012,fd=4),("apach
e2",pid=1011,fd=4),("apache2",pid=1010,fd=4),("apache2",pid=1009,fd=4),("apache2",pid=1008,fd=4),("
apache2",pid=966,fd=4))

```

```

(kali㉿kali)-[~/CIP-B103-Lab1]
$ tree ~/CIP-B103-Lab1
/home/kali/CIP-B103-Lab1
├── evidence
│   ├── basic.pcapng
│   └── root_test
├── exported
├── reports
│   ├── basic_pcap_sha256.txt
│   └── curl_verbose.txt
├── screenshots
├── scripts
└── working

7 directories, 4 files

(kali㉿kali)-[~/CIP-B103-Lab1]
$ █

```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo tshark -i lo -f 'tcp port 80' -w /tmp/basic.pc
Running as user "root" and group "root". This could be
Capturing on 'Loopback: lo'
10 ^C

(kali㉿kali)-[~/CIP-B103-Lab1]
$ sudo cp /tmp/basic.pcapng evidence/basic.pcapng
sudo chown kali:kali evidence/basic.pcapng

(kali㉿kali)-[~/CIP-B103-Lab1]
$ ls -l evidence/basic.pcapng
-rw----- 1 kali kali 1828 Jul 28 08:23 evidence/basic
.pcapng

(kali㉿kali)-[~/CIP-B103-Lab1]
$ cp --preserve=timestamps evidence/basic.pcapng work
ing/basic_working.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ curl --no-keepalive -v http://127.0.0.1/basic.html
* Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 8
0) from 127.0.0.1 port 40264
* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 28 Jul 2026 12:23:06 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Tue, 28 Jul 2026 11:50:25 GMT
< ETag: "78-657aa71cb8292"
< Accept-Ranges: bytes
< Content-Length: 120
< Vary: Accept-Encoding
< Content-Type: text/html
```

```
<!DOCTYPE html>
<html><body><h1>CIP-B103 HTTP Evidence</h1><p>Name: Fus
eini Imoru</p><p>Reg No: 17291</p></body></html>
* Connection #0 to host 127.0.0.1:80 left intact
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ sha256sum evidence/basic.pcapng working/basic_worki
ng.pcapng | tee reports/capture_hashes.txt
2e8b38d4cddf38b0e338f689287b45ebfc4ff939731da83e29fbfe5
ea162617e evidence/basic.pcapng
2e8b38d4cddf38b0e338f689287b45ebfc4ff939731da83e29fbfe5
ea162617e working/basic_working.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ tshark -r working/basic_working.pcapng -Y 'tcp.stream eq 0' \
-T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e tcp.flags -e tcp.seq -e tcp.ack | tee reports/full_handshake.tsv
1 2026-07-28T08:23:06.532514572-0400 127.0.0.1 40264 127.0.0.1 80 0x0002 0 0
2 2026-07-28T08:23:06.532577563-0400 127.0.0.1 80 127.0.0.1 40264 0x0012 0 1
3 2026-07-28T08:23:06.532590405-0400 127.0.0.1 40264 127.0.0.1 80 0x0010 1 1
4 2026-07-28T08:23:06.532709748-0400 127.0.0.1 40264 127.0.0.1 80 0x0018 1 1
5 2026-07-28T08:23:06.532781123-0400 127.0.0.1 80 127.0.0.1 40264 0x0010 1 84
6 2026-07-28T08:23:06.541823214-0400 127.0.0.1 80 127.0.0.1 40264 0x0018 1 84
7 2026-07-28T08:23:06.541874437-0400 127.0.0.1 40264 127.0.0.1 80 0x0010 84 372
8 2026-07-28T08:23:06.542440768-0400 127.0.0.1 40264 127.0.0.1 80 0x0011 84 372
9 2026-07-28T08:23:06.542800566-0400 127.0.0.1 80 127.0.0.1 40264 0x0011 372 85
10 2026-07-28T08:23:06.542821088-0400 127.0.0.1 40264 127.0.0.1 80 0x0010 85 373
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ tshark -r working/basic_working.pcapng -Y 'http.request' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e http.request.method -e http.request.uri -e http.host -e http.user_agent \
| tee reports/http_requests.tsv
4 2026-07-28T08:23:06.532709748-0400 127.0.0.1 40264 127.0.0.1 80 GET /basic.html 1
27.0.0.1 curl/8.20.0
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$ tshark -r working/basic_working.pcapng -Y 'http.response' -T fields \
-e frame.number -e frame.time -e http.response.code -e http.response.phrase \
-e http.server -e http.content_type -e http.content_length \
| tee reports/http_responses.tsv
6 2026-07-28T08:23:06.541823214-0400 200 OK Apache/2.4.68 (Debian) text/html 120
```

```
(kali㉿kali)-[~/CIP-B103-Lab1]
$
```


